
IAL Biology Unit 4

Rapid Revision

WBI14 — Energy, Environment, Microbiology and Immunity

Last-minute cram sheets — designed to be scanned, not read

Condensed from the full A* study note
(21 past papers and mark schemes, June 2019 - June 2026)

90 marks · 1 h 45 min · 8–9 questions, no sections · 1–3 asterisked (*) levels questions

A* rule: grade A overall ($\geq 480/600$ UMS) **and** $\geq 90\%$ across Units 4, 5 and 6 combined ($\geq 270/300$). The A* is decided entirely by A2 performance.

TRIAGE — WHAT THIS PAPER REWARDS

WHERE THE MARKS ARE

Photosynthesis (12–20) · immunity (10–18) · microbiology (10–18) · climate change (8–16) · energy flow (6–14) — **every paper**. Prepare these five plus evolution and forensics and you have 70–80 of the 90 marks.

WHAT THE MARK SCHEMES REWARD THAT STUDENTS MISS

1. Use the examiner's exact word — the schemes list rejections. **2.** Linked chains, not lists: six unlinked facts score 4, not 6. **3.** Quote figures with units. **4.** On every * question, pair each description with its explanation — describing alone caps you at Level 1–2.

UNIT 4 — SHEET 1 · PHOTOSYNTHESIS (every paper, 12–20 marks)

Light-dependent → **thylakoid membranes**. **Light-independent (Calvin)** → **stroma**. $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow (\text{light, chlorophyll}) \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$

THE LDR CHAIN — SIX STEPS

1. Light absorbed by chlorophyll in the photosystems · **2.** electrons excited and released · **3.** electrons pass along the electron transport chain, releasing energy · **4.** energy pumps protons into the **thylakoid space**, creating a proton gradient · **5.** protons diffuse back through **ATP synthase** — chemiosmosis — making ATP · **6.** NADP accepts electrons and protons → **reduced NADP**.

Photolysis: $2\text{H}_2\text{O} \rightarrow 4\text{H}^+ + 4\text{e}^- + \text{O}_2$ — replaces electrons lost by chlorophyll, provides protons for reduced NADP, releases oxygen.

LEARN VERBATIM — THE 2-MARK PIGMENT ANSWER (asked 3+ times)

"to absorb light **so that the electrons will be excited / released from photosystems**" · "so that reduced NADP and ATP can be synthesised"

Calvin cycle	What happens
1. Carbon fixation	$\text{CO}_2 + \text{RuBP (5C)} \rightarrow [\text{RUBISCO}] \rightarrow 2 \times \text{GP (3C)}$
2. Reduction	$\text{GP} + \text{reduced NADP} + \text{ATP} \rightarrow \text{GALP (3C)}$
3. Regeneration	5 of every 6 GALP + ATP → regenerate RuBP; 1 in 6 leaves

Per hexose: **6 turns · 6 CO₂ · 18 ATP · 12 reduced NADP**.

RATE CHANGES — THE 5-LINK CHAIN (3 marks awarded)

more light → **more photolysis** → more electrons released → **more ATP and reduced NADP** → Calvin cycle faster / **RuBP regenerated faster to bind more CO₂**.

Reverse (net CO₂ uptake negative): respiration > photosynthesis · **RuBP regenerated too slowly to bind as much CO₂** · light intensity very low. "DO NOT ACCEPT no photosynthesis." The RuBP clause is the discriminating mark.

Chloroplast: double membrane · thylakoids in grana (photosystems, pigments, ATP synthase, large SA) · **small thylakoid space** so the proton gradient builds quickly · stroma (RUBISCO, Calvin enzymes) · **DNA loop** · **70S ribosomes** · starch grain.

GALP → : glucose → starch/cellulose/sucrose (condensation, glycosidic) · glycerol + fatty acids (ester bonds) · **amino acids from glucose + nitrates** (glucose has no nitrogen).

Absorption spectrum = light absorbed per wavelength; **action spectrum** = rate of photosynthesis per wavelength. Similar shape → the pigments absorbing light drive photosynthesis.

Rf = distance moved by pigment ÷ distance moved by solvent front. Order: carotene (orange, highest) · xanthophyll · chlorophyll a · chlorophyll b (lowest). Carotene travels furthest — **most soluble in the solvent**.

Rate practical: Elodea/Cabomba in hydrogencarbonate · **photosynthometer** beats bubble counting · **heat shield** so only light intensity changes · 5 min to equilibrate · gas is not pure O₂ and some dissolves, so volume underestimates.

UNIT 4 — SHEET 2 · ENERGY FLOW & ECOLOGY

GPP = total energy fixed by photosynthesis per unit area per unit time · **R** = energy used in respiration · **NPP = GPP – R**. Units **kJ m⁻² yr⁻¹** — must contain an **area and a time**. In a table **GPP is always largest**.
efficiency (%) = (energy in higher level ÷ energy in lower) × 100

Why so little transfers: not all eaten (bones, roots, bark) · not all digested — egested in faeces · heat lost in respiration (especially endotherms) · lost in excretion (urea). Consequences: food chains rarely exceed 4–5 links; eating plants is more efficient than meat.

Term	The accepted definition — with its rejection
Habitat	The place where an organism lives
Population	All the organisms of one species in a habitat at one time
Community	All the populations of different species living and interacting in a habitat
Ecosystem	Organisms interacting with each other AND with the abiotic / physical environment — "IGNORE habitat". Both halves needed.
Niche	" The role that an organism plays " — plus a stated reason when the question gives a context (food source, temperature tolerance, desiccation)
Biodiversity	Number of different species (species richness) + genetic diversity within a species — two components, two marks

Sampling: quadrats for abundance (random coordinates to avoid bias; density, frequency or % cover; species-area curve to find enough samples). **Transects** for distribution along a gradient. Population = mean per quadrat × (total area ÷ quadrat area).

Succession: pioneers colonise bare rock → die and decompose → **humus and thin soil** → soil holds water and nutrients → new species outcompete pioneers → **biodiversity increases at each sere** → stable **climax community**. Secondary succession is faster — soil, nutrients and a seed bank already present.

UNIT 4 — SHEET 3 · CLIMATE CHANGE

Evidence	Mechanism	Limitation
CO₂ records	Direct measurement; air bubbles in ice cores dated by annual layers	Polar regions only; dating uncertain at depth
Temperature	Thermometers (~150 yr); before that ¹⁸O:¹⁶O ratios in ice cores — more ¹⁸ O = warmer	Short record; uneven stations; urban heat islands
Pollen in peat bogs	Peat is acidic and anaerobic → slow decomposition → pollen preserved; species-specific shapes; deeper = older; carbon dating	Pollen may blow in; not all plants make durable pollen; tolerances may have changed
Dendrochronology	One ring per year; wider = warmer/wetter; overlapping samples extend the record	Width also depends on nutrients, disease, competition, shading

"ANTHROPOGENIC CLIMATE CHANGE" — 2 MARKS, 2 CLAUSES

"caused by the **effects of humans**" — "**IGNORE named activities**" · "changes to **mean temperature / rainfall**" — "ACCEPT long-term change in weather patterns"; "**IGNORE weather unqualified**".
Listing "burning fossil fuels" earns **nothing**.

Greenhouse effect — 4 steps: short-wavelength radiation passes through and is absorbed by the surface → surface **re-radiates at longer wavelengths (infrared)** → greenhouse gases absorb infrared and re-radiate some back → less energy escapes, mean surface temperature rises. Gases: CO₂, methane, water vapour, nitrous oxide, CFCs. **Oxygen and nitrogen are not.**

Methane chain: microorganisms secrete enzymes → break down organic matter → sugars respired → **anaerobic conditions produce methane** → methane **traps infrared** → increase in temperature of **the atmosphere / surface / water** — "**IGNORE global temperatures**".

Effects on organisms — three categories: **distribution** (move polewards / to altitude; those at poles or mountain tops have nowhere to go) · **development** (most are ectothermic → faster enzyme-controlled development up to the optimum) · **life cycles** (earlier flowering, egg-laying, emergence → risk of **desynchronisation** with food supply).

Q₁₀ = rate at (T + 10 °C) ÷ rate at T — **NO UNITS** ("DO NOT ACCEPT with units"), and only meaningful below the optimum.

Why a prediction is unreliable: data fluctuates · factors increasing or decreasing warming could change · **the trend could change** · older data less accurate.

UNIT 4 — SHEET 4 · EVOLUTION & SPECIATION

THE 3-MARK NATURAL SELECTION CHAIN — AND ITS TWO FATAL REJECTIONS

mutation (in the DNA) causing an **allele for resistance** · the chemical acts as a **selection pressure** · **resistant** individuals survive and **pass the allele to their offspring**.

"**DO NOT ACCEPT immune for resistant.**" "**DO NOT ACCEPT selection pressure causes mutation.**" Mutation is random and comes **first**.

Why resistance evolves fast: short life cycle · high mutation rate · many offspring · strong selection pressure. Bacteria also spread it **horizontally by conjugation via plasmids**.

Evolution = a change in allele frequency in a population.

Allopatric = geographical barrier. **Sympatric** = same area, ecological/behavioural/temporal isolation. The full chain: **reduced gene flow → changes in allele frequency → reproductive isolation → two species**.

Read which is asked. "How did X evolve feature Y" (selection chain) sits next to "how did X become a new species" (speciation chain). Different questions, different marks.

UNIT 4 — SHEET 5 · MICROBIOLOGY

"DESCRIBE HOW BACTERIA CAN BE GROWN" — 3 OF 4 POINTS

on agar / in a broth / in nutrients — "IGNORE grown in a Petri dish" · **aseptic technique** (or a description of it) · **incubated at a stated temperature in the range 25-35 °C** · another culture detail (pH, growth factors, 24 h-1 week, not anaerobic).

"In a Petri dish at a suitable temperature" scores **zero** of those two marks. Name the medium, state a number.

Aseptic technique: autoclave (121 °C, 15 min) · disinfect the surface · work **close to a Bunsen flame** (updraught) · flame the loop before and after · flame bottle necks · lift the lid **slightly and at an angle** · tape but **do not seal** — sealing creates **anaerobic** conditions for harmful anaerobes · incubate **inverted at ≤25 °C** in schools · autoclave before disposal.

Growth measurement	Advantage	Disadvantage
Cell count (haemocytometer)	Quick; absolute number	Counts living and dead
Dilution plating (viable count)	Counts only living ; detects low numbers	Assumes 1 colony = 1 cell; 24-48 h
Dry mass to constant mass	Direct measure of biomass	Destructive; slow
Turbidity / optical density	Quick, non-destructive, continuous	Counts dead cells and debris; needs a calibration curve

cells per cm³ = (colonies ÷ volume plated) × dilution factor. Use a plate with **30-300 colonies** — below 30 is unreliable, above 300 colonies merge.

Why one calibration curve won't serve all species: different **cell sizes, shapes and pigmentation** scatter and absorb light differently, so the same optical density means a different cell number.

GROWTH CURVE — FOUR PHASES

Lag — cells synthesising enzymes, taking up water and nutrients; little division · **Log** — nutrients and space plentiful, binary fission at maximum rate, population doubles at a constant interval · **Stationary** — division rate = death rate; nutrients limiting, toxic waste, lack of O₂ and space · **Death** — death rate exceeds division.

k = (log N_t - log N₀) ÷ (0.301 t) — values **must be taken from the log/exponential phase** (that is a mark in itself). Also $N = N_0 \times 2^n$; generation time = $t \div n$.

Why a log scale: numbers span orders of magnitude, and the exponential phase becomes a straight line whose gradient gives the growth rate.

	Bacteria	Viruses
Cell?	Prokaryotic cell	Not a cell
Genetic material	Circular DNA + plasmids	DNA or RNA
Coat / wall	Peptidoglycan cell wall	Capsid ; some have an envelope with glycoproteins
Ribosomes	70S	None
Reproduction	Binary fission, independently	Only inside a host cell

Why viruses need living cells: they cannot replicate without a host · they need the host's **enzymes, monomers and ribosomes** — "**DO NOT ACCEPT to grow**".

Why different viruses go to different parts of an egg: each has a specific host cell and **attachment molecules / receptors** — "**IGNORE antigens**" throughout this answer.

The four named viruses: Ebola (RNA, helical, enveloped, humans/primates) · TMV (RNA, helical rod, no envelope, plants) · HIV (RNA + reverse transcriptase, complex, gp120 envelope, **T helper cells**) · λ phage (DNA, head+tail+fibres, *E. coli*).

Lytic: attachment → entry → replication → assembly → lysis. **Latency:** viral nucleic acid integrated into host DNA, replicated with it, no new particles until a trigger.

UNIT 4 — SHEET 6 · IMMUNITY

THE SINGLE MOST-PENALISED ERROR IN THE TOPIC

Macrophages present antigen **to T helper (CD4) cells** — "**DO NOT ACCEPT to T killer cells / B cells / other cells.**"

The specific response — 8 steps: macrophage engulfs pathogen and presents antigen on MHC → presented to a **T helper** cell with a complementary receptor → T helper activated, divides by mitosis, releases **cytokines** → cytokines activate B cells and T killer cells → B cell divides into **plasma cells and B memory cells** → plasma cells secrete **antibodies** → T killer cells destroy infected cells with **perforin** → memory cells persist.

THE PRIMARY-RESPONSE DELAY — 4 MARKS, WITH A LINKAGE RULE

"**NB answer must be linked with a time delay / many-staged process / slow process** for max 4 marks to be awarded."

first infection so **no memory cells** · time for macrophages to bind, engulf and process · time to present antigen to T helper cells · time for T helper cells to activate B cells · time for B cells to **differentiate into plasma cells** · time for plasma cells to **synthesise antibody**.

A list of stages **without the time framing** is capped.

Phagocytosis: chemotaxis → recognises non-self via receptors (opsonins) → engulfs by endocytosis into a **phagosome** → **lysosome fuses** → hydrolytic enzymes digest → presents the antigen.

Inflammation: mast cells release **histamine** → **vasodilation of arterioles** and increased capillary permeability → white blood cells and antibodies brought to the site. "**IGNORE capillaries dilate**" — say blood flow increases / vasodilation. Four signs: redness, heat, swelling, pain.

Gut flora: reduce growth of pathogens because they **compete for nutrients or space** — "**IGNORE food**"; also produce vitamin K.

Antibodies: Y-shaped, 4 chains, disulfide bonds, **two variable regions**, constant region, hinge. Act by **agglutination** · **neutralisation** · **opsonisation** · **complement activation**.

Antigenic variability (flu, HIV) · hiding in host cells (TB in macrophages) · **preventing lysosome fusion** (TB waxy wall) · latency · destroying immune cells (HIV) · suppressing the response (Ebola VP35 inhibits interferon) · capsules.

	Active	Passive
Natural	Infection	Antibodies from the mother — placenta, breast milk
Artificial	Vaccination	Injection of antibodies (antiserum)
Memory cells	Yes — long-lasting	No — immediate but short-lived

THE ANTIBIOTIC DEFINITION — 2 MARKS, THREE TRAPS

"a **substance / chemical / drug**" — "**DO NOT ACCEPT produced by a virus**" · "that **kills or inhibits the growth of bacteria**" — "**DO NOT ACCEPT virus**"; "**IGNORE pathogens**".

Bacteriostatic (inhibits; graph plateaus; tetracycline) vs **bactericidal** (kills; graph falls; penicillin). They target **peptidoglycan cell walls, 70S ribosomes, DNA gyrase, folate synthesis** — human cells have no wall and 80S ribosomes; viruses have no wall, ribosomes or metabolism.

Herd immunity: vaccinated people will not develop the disease · **fewer infected people to infect those who are not immune** · protects those who cannot be vaccinated (allergic, immunodeficient).

UNIT 4 — SHEET 7 · FORENSICS, DECOMPOSITION, PCR

Method	Basis	Limitation — where the marks are
Body temperature (algor mortis)	Metabolism stops → cools towards ambient, ~1.5–2 °C per hour, sigmoid curve; core temperature measured	Useless once ambient is reached (it has levelled off). Cooling depends on ambient temperature, body size/SA:V, fat, clothing, wind, immersion. Cooler environment → shorter estimate , because heat is lost faster.
Rigor mortis	No ATP → actin-myosin cross-bridges cannot detach → muscles stay contracted. Small muscles first at 4–6 h, complete ~12 h, passes off 24–36 h as enzymes break down the proteins	Temperature / exercise / body fat affect it · deciding when a body is stiff is subjective · the same stiffness occurs twice — during onset and release
Decomposition	Autolysis then putrefaction — gases, bloating, discolouration, skin slippage, skeletonisation	Depends on temperature, oxygen, water, location, and which decomposers colonise
Forensic entomology	Blowflies arrive within minutes–hours, lay eggs in moist areas; larval instars of known size take known times at known temperatures	Gives only a MINIMUM time since death . Don't know how far into a stage · other factors affect development · body may have been moved · insects may have been excluded (indoors, buried, wrapped)

Insect succession: species appear **in a specific sequence** — *Lucilia* first (pioneer), then *Cochliomyia/Chrysomya* as *Lucilia* is outcompeted, then *Ophyra*.

The evaluation answer: any single method gives only a range; assumptions may not hold; use several methods together, each accurate over a different window, and give a range.

Saprotrophic digestion: enzymes **packaged by the Golgi into vesicles, released by exocytosis** → hydrolysis of polymers **outside** the organism → soluble products absorbed → respired, releasing CO₂. Fungi: **hyphae give a large surface area**. Detritivores increase surface area for decomposers.

PCR — AND THREE WAYS TO LOSE THE REAGENT MARK

95 °C denaturation (hydrogen bonds break) → **50–65 °C annealing** (primers bind) → **72 °C extension** (**Taq** DNA polymerase adds nucleotides).

Reagents: primers · **mononucleotides** — "IGNORE bases", "DO NOT ACCEPT RNA nucleotides" · **Taq / DNA polymerase** — "DO NOT ACCEPT RNA polymerase" · buffer.

Why Taq: from a thermophile, so not denatured at 95 °C. **copies = 2ⁿ × starting number**.

Electrophoresis: cut with **restriction endonucleases** → load into agarose wells → DNA is **negatively charged (phosphate groups)** so moves to the **anode** → **shorter fragments move further** (less resistance through the pores) → stain or probe, size against a ladder. Higher agarose concentration = smaller pores = better separation of short fragments. Circular DNA moves faster (compact/supercoiled). Profiling uses **non-coding repeated sequences** because there is **no selection pressure keeping them constant**.

THE REJECTIONS BANK — THE MARKS MOST OFTEN LOST

Do not write	Write instead	
	"Immune" (to a chemical or antibiotic)	Resistant
	"The selection pressure caused the mutation"	Mutation is random and occurs first
	Antigen presented to T killer cells / B cells	T helper (CD4) cells
	"Grown in a Petri dish at a suitable temperature"	On agar / in broth, at a stated 25-35 °C
	Antibiotics produced by a virus, or acting on viruses; "pathogens"	Produced by microorganisms; act on bacteria
	"Capillaries dilate"	Vasodilation / blood flow increases
	Gut flora "compete for food"	Compete for nutrients or space
	Oxygen in glucose comes from water	C and O from CO₂ ; H from water
	Ecosystem without the abiotic environment; "habitat"	Organisms and the abiotic environment
	Q ₁₀ given with units; "global temperatures"	No units; atmosphere / surface / water
	PCR: "bases", "RNA nucleotides", "RNA polymerase"	Mononucleotides; Taq / DNA polymerase
	Viruses need host cells "to grow"	They need the host's enzymes, monomers and ribosomes

WALKING INTO THE EXAM

Unit 4 — circle the command word and the mark allocation; on every * question alternate description and explanation; quote figures with units; use the examiner's exact word.